Dev Patel

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Professional Summary

Detail-oriented **Data Scientist and Machine Learning Engineer** with a strong foundation in **Predictive Modeling, Time Series Analysis, Computer Vision, and Geospatial Analytics**. Proficient in leveraging **Python, PyTorch, XGBoost, and Scikit-learn** to extract actionable insights from complex datasets. Demonstrated success in building end-to-end ML pipelines, deploying interactive analytics dashboards, and optimizing models for real-world automation and business intelligence.

Technical Skills

Machine Learning & Deep Learning: XGBoost, LightGBM, Random Forest, SVM, YOLOv12, PyTorch, Scikit-learn
Data Science & Analytics: Time Series Forecasting, NLP (TF-IDF, BoW), GeoPandas, OSMnx, Pandas, NumPy, SciPy
Data Engineering & Deployment: SQL, PostgreSQL, Streamlit, FastAPI, Docker, Git/GitHub, Cloud Deployment

Professional Experience

NielsenIQ (NIQ) Vadodara, Gujarat
Data Science Intern *Nov 2025 – May 2026*

- Engineered an end-to-end ML pipeline (GeoRoadMetrics) to predict urban road quality across 2.1 million geospatial segments, deploying an XGBoost classifier that achieved **85.88% accuracy** and an F1-score of 0.831.
- Developed a scalable data processing engine using Python, GeoPandas, and OSMnx to automate road network extraction and UTM-projected geometric buffering, achieving **95% precision** in surface area calculations.
- Architected an interactive **Streamlit web dashboard** integrating Folium to visualize predictive analytics, statistical anomaly metrics, and dynamic geospatial mapping.

BrainyBeans Ahmedabad, Gujarat
Data Science Intern *May 2025 – Jun 2025*

- Built and optimized regression models for smart-building infrastructure forecasting, leveraging Scikit-learn to achieve an **R^2 score of 0.82**.
- Authored data preprocessing scripts to execute rigorous feature engineering and optimize model hyperparameters.

Key Projects

Brain Tumor Detection using YOLOv12 GitHub
Computer Vision, Deep Learning, PyTorch, Medical Imaging

- Trained and fine-tuned a custom **YOLOv12 object detection model** on MRI scans, achieving a benchmark **mAP@0.5 of 91.39%** (outperforming baseline ResNet18 by 27%).
- Implemented advanced image augmentation techniques (Mosaic, MixUp) to successfully mitigate severe class imbalances typical in medical imaging datasets.

Historical Price Forecasting & Time Series Analysis GitHub
Python, Pandas, Scikit-learn, SciPy, Matplotlib

- Executed comprehensive time-series analysis on 20 years of historical monthly price data (2005-2025), handling complex temporal patterns and non-linear growth trends.
- Implemented rigorous outlier detection (IQR and Z-Score algorithms) to isolate significant market anomalies, identifying critical volatility periods featuring ¿90% price spikes.

Fake News Detection Analytics Platform GitHub
Streamlit, NLP, Random Forest, SVM, Naive Bayes

- Built an interactive **Streamlit NLP dashboard** evaluating multiple vectorizers (TF-IDF, BoW) across 5 distinct classification models, delivering real-time performance metrics, feature importance, and precision-recall diagnostics.

Education

Charotar University of Science and Technology (CHARUSAT) Gujarat, India
B.Tech in Computer Science and Engineering — CGPA: 8.63 *Expected May 2026*

Publications & Certifications

- **Publication:** *"Enhancing Brain Tumor Detection Using YOLOv12"* – Published in IEEE 3rd International Conference on Inventive Computing and Informatics (ICICI), 2025. View Paper
- **Google Advanced Data Analytics** – Professional Certificate
- **Microsoft AI and ML Engineering** – Professional Certificate